UNIVERSITY OF CAPE COAST
COLLEGE OF AGRICULTURAL AND NATURAL SCIENCES
SCHOOL OF PHYSICAL SCIENCES
DEPARTMENT OF PHYSICS
COURSE OUTLINE FOR FIRST SEMESTER, 2022/2023 ACADEMIC YEAR
JANUARY, 2023

Course Code and Title: PHY 101 – GENERAL PHYSICS 1

Instructor Information	**Teaching Assistant Information**
Augustine Arthur University of Cape Coast College of Agricultural and Natural Sciences School of Physical Sciences Department of Physics Tel. No: 0207340766 WhatsApp: 0207340766 Telegram: 0207340766 Official Email: aarthur@ucc.edu.gh Academic Email: aarthur.ucc@gmail.com	Miss Hawa Jibril Tel No: 0558517618 WhatsApp: 0558517618 Telegram: 0558517618 Official Email: jibrilh9035@gmail.com

Teaching/Tutorial:
10 weeks X > 2 hours of lectures / tutorials = 36 hours

Lecture Time and Venue:
Thurs, 12:30-2:30pm, LT17

Tutorial Times
xxx, xxx

Grading/Evaluation: The course interactions consists of 10 lectures, 2 assessment quizzes, 2 assessment assignments. A mini project and/or credits for attendance and class participation. Some of the lectures will require you to download files from /submit files to the internet. All documents are in MS WORD/PDF/.ppt format.
Minimum of 2 quizzes, 2 assignments and mini project = 40% (CONTINOUS ASSESSMENT)
End of semester examinations = 60% (FINAL)
Mode of Delivery: Face to Face and online (Google meet, zoom).

Course Learning Outcome:

Course length and Pace:	**Academic Integrity:**
This course is delivered over 10-week period with two hours of lectures each week	Students may discuss their assignments with each other and work together on them when deemed appropriate. Each individual is expected to submit their own individual assignment you are assigned in group or groups to do group work for mini projects

Course Materials:

Reading Lists

General feedback: Student input is very much welcomed.

Lecture Plan

WEEK	TOPICS / DESCRIPTIONS	REMARKS
1	**Introduction to mechanics** ➢ Fundamental and Derived Quantities, Standard Prefixes, Unit Conversions ➢ Dimensional Analysis Physical Quantities & Units	
2 & 3	**Scalars and vectors** ➢ Components of a vector ➢ Unit vectors ➢ Addition and subtraction of vectors ➢ Multiplication of vectors Vectors I: addition, graphical resultants, resolution Vectors II: unit vectors, dot and cross products	Assignment
4 & 5	**Dynamics** ➢ Force and types of forces ➢ Newton's laws of motion ➢ Impulse and momentum ➢ Inelastic and elastic collisions ➢ Strings, pulleys and inclined planes Dynamics: force, mass, weight, Newton's laws & their applications Dynamics: Momentum; Types of collisions; friction	Quiz 1
6 & 7	**Kinematics** ➢ Displacement, speed, velocity and acceleration ➢ Graphs of motion ➢ Circular motion ➢ Simple harmonic motion ➢ Projectile motion	

	Kinematics: graphs, equations of motion, projectiles Circular Motion, Simple Harmonic Motion	
8	**Work, Energy and Power** Mechanical Work, Energy and Power	Quiz 2
9	**Elasticity** ➢ Stress and string ➢ Young's modulus ➢ Shear modulus ➢ Bulk modulus ➢ Hooke's law Elasticity: Young's modulus; stress-strain curve	
10	**Heat energy** ➢ Kinetic theory of gases ➢ Temperature scales ➢ Lab and clinical thermometers compared ➢ Heat capacities of matter ➢ Thermal conductivity Kinetic theory and gas laws; Thermometry Heat Capacity & Thermal Conductivity	
11-13	**Revision & Exams**	

REFERENCES

1. Halliday, D., Resnick, R., & Walker, J. (2013). *Fundamentals of physics*. John Wiley & Sons.
2. Serway, R. A., & Jewett, J. W. (2018). *Physics for scientists and engineers*. Cengage learning.
3. Cutnell, J. D., & Johnson, K. W. (2014). *Physics, Volume One: Chapters 1-17* (Vol. 1). John Wiley & Sons.